SESSION 2

Problem Statement Document

AI Career Guidance & Skill Recommendation System

Project	AI Career Guidance & Skill Recommendation System
Deliverable	Problem Statement Document
Industry Mapping	Product Requirement Analysis
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Repository	[Add GitHub repository link]

Topics Covered in This Session

- Identification of real-world software problems
- Problem statement writing
- Technical and operational feasibility
- Project scope definition

1. Real-World Problem Identification

Every year lakhs of engineering students graduate with skills that do not match what employers need. Students choose careers based on peer pressure, scattered YouTube advice or trends, rather than a structured comparison between their own profile and the requirements of a role.

Observed problem	Effect on students
No personalised guidance on which role suits their skills	Applying to wrong roles; low selection rates
Skill gaps are discovered late (during placements)	Little time left to learn missing skills
Learning resources are overwhelming and unstructured	Wasted effort, no clear order of learning
College counselling does not scale to every student	Many students get no guidance at all
No visibility of which skills are in demand	Learning outdated or low-demand technologies

2. Problem Statement

Students lack an affordable, personalised and data-driven way to understand which careers suit their current skills, what skills they are missing, and what to learn next. A web-based system is needed that reads a student's resume, matches the profile with career requirements, identifies skill gaps, and produces a step-by-step learning roadmap with course suggestions, supported by an AI chatbot for ongoing questions.

Problem statement in 5W format

Who	Engineering and degree students; secondary users are placement cells and admins
What	Lack of personalised career direction and clarity on skill gaps
Where	On a web platform accessible from any device
When	During the semesters before placements and internships
Why	To improve employability and reduce guesswork in career planning

3. Objectives

1. Extract skills, education, projects and experience from an uploaded resume using Claude AI.
2. Recommend suitable career roles with a transparent match score.
3. Identify skills the student has and skills still missing for the chosen role.
4. Generate a personalised learning roadmap ordered by priority.
5. Recommend courses and resources for every skill gap.
6. Provide an analytics dashboard of in-demand skills, popular careers and common gaps.
7. Offer an AI career chatbot for questions about careers, skills and preparation.

4. Existing Solutions and Gaps

Existing approach	Limitation
Generic career blogs and videos	Not personalised; no skill gap measurement
Online course platforms	Recommend courses by popularity, not by the student's gap
Job portals (resume scoring)	Focus on one job posting, no learning plan
Manual counselling	Expensive, limited availability, not scalable

Our differentiator: one platform that connects resume analysis, career matching, gap analysis, roadmap, courses and chat in a single flow.

5. Feasibility Analysis

Type	Assessment	Verdict
Technical	- MERN is mature and well documented • Claude API handles resume understanding and roadmap generation • Pandas covers analytics • Main risk: parsing quality of varied resume layouts, mitigated by text extraction plus AI structuring	Feasible
Operational	- Students already use web apps; no special hardware • Simple flow: upload, view results, chat • Admin can update careers and courses without code changes • Needs moderation of AI output for accuracy	Feasible
Economic	- Free tiers for MongoDB Atlas and hosting • Claude API is pay per use; cost controlled by caching results and limiting requests per user	Feasible (low cost)
Schedule	- Scope split into sprints across one semester • Core flow (resume to roadmap) first, analytics and chatbot after	Feasible
Legal / Ethical	- Resumes contain personal data: store securely, collect consent, allow deletion • Advice shown as guidance, not a guarantee	Manageable

6. Project Scope

In scope

- User registration and login
- Resume upload (PDF/DOCX) and AI analysis
- Career recommendation with match score
- Skill gap analysis
- AI roadmap generation
- Course recommendation
- Analytics dashboard
- AI career chatbot
- Admin management of careers and courses

Out of scope (this version)

- Live job listings and direct job applications
- Video interviews and mock tests
- Payments or paid mentoring
- Native mobile apps
- Automatic course enrolment on third-party sites
- Languages other than English

Assumptions and constraints

- Users have a resume in PDF or DOCX format and internet access.
- Claude API availability and rate limits affect response time.
- Career requirement data is maintained by admins and treated as the source of truth.
- Project is delivered by a student team within the semester timeline.

7. Risks and Mitigation

Risk	Impact	Mitigation
Resume formats vary widely	Wrong or missing skills	Extract text first, ask AI for structured JSON, let user edit extracted skills
AI gives inaccurate advice	Loss of trust	Ground answers in stored career data; show disclaimer
API cost or downtime	Slow or failed analysis	Cache results, retry logic, rate limits
Personal data leakage	Privacy violation	JWT auth, HTTPS, access control, delete-my-data option
Scope creep	Delays	Fixed scope list; new ideas go to backlog

8. Success Criteria

Measure	Target
Resume analysis completes	Within 15 seconds for a 2-page resume
Skill extraction accuracy (manual check on sample resumes)	85% or higher
Every recommendation shows a match score and missing skills	100% of results
User satisfaction in pilot survey	4 out of 5 or higher